

From Words to Worlds: Spatial Intelligence is AI's Next Frontier



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In 1950, when computing was little more than automated arithmetic and simple logic, Alan Turing asked a question that still reverberates today: can machines think? It took remarkable imagination to see what he saw: that intelligence might someday be built rather than born. That insight later launched a relentless scientific quest called Artificial Intelligence (AI). Twenty-five years into my own career in AI, I still find myself inspired by Turing's vision. But how close are we? The answer isn't simple.

Today, leading AI technology such as large language models (LLMs) have begun to transform how we access and work with abstract knowledge. Yet they remain wordsmiths in the dark; eloquent but inexperienced, knowledgeable but ungrounded. **Spatial intelligence will transform how we create and interact with real and virtual worlds—revolutionizing storytelling, creativity, robotics, scientific discovery, and beyond. This is AI's next frontier.**

The pursuit of visual and spatial intelligence has been the North Star guiding me since I entered the field. It's why I spent years building ImageNet, the first large-scale visual learning and benchmarking dataset and one of three key elements enabling the birth of modern AI, along with neural network algorithms and modern compute like graphics processing units (GPUs). It's why my academic lab at Stanford has spent the last decade combining computer vision with robotic learning. And it's why my cofounders Justin Johnson, Christoph Lassner, Ben

Mildenhall, and I created [World Labs](#) more than one year ago: to realize this possibility in full, for the first time.

In this essay, I'll explain what spatial intelligence is, why it matters, and how we're building the world models that will unlock it—with impact that will reshape creativity, embodied intelligence, and human progress.

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Spatial Intelligence: The scaffolding of human cognition

AI has never been more exciting. Generative AI models such as LLMs have moved from research labs to everyday life, becoming tools of creativity, productivity, and communication for billions of people. They have demonstrated capabilities once thought impossible, producing coherent text, mountains of code, photorealistic images, and even short video clips with ease. It's no longer a question of whether AI will change the world. By any reasonable definition, it already has.

Yet so much still lies beyond our reach. The vision of autonomous robots remains intriguing but speculative, far from the fixtures of daily life that futurists have long promised. The dream of massively accelerated research in fields like disease curation, new material discovery, and particle physics remains largely unfulfilled. And the promise of AI that truly understands and empowers human creators—whether students learning intricate concepts in molecular chemistry, architects visualizing spaces, filmmakers building worlds, or anyone seeking fully immersive virtual experiences—remains beyond reach.

To learn why these capabilities remain elusive, we need to examine how spatial intelligence evolved, and how it shapes our understanding of the world.

Vision has long been a cornerstone of human intelligence, but its power emerged from something even more fundamental. Long before animals could nest, care for their young, communicate with language, or build civilizations, the simple act of sensing quietly sparked an evolutionary journey toward intelligence.

This seemingly isolated ability to glean information from the external world, whether a glimmer of light or the feeling of texture, created a bridge between perception and survival that only grew stronger and more elaborate as the generations passed. Layer upon layer of neurons grew from that bridge, forming nervous systems that interpret the world and coordinate interactions between an organism and its surroundings. Thus, many scientists have conjectured that **perception and action became the core loop driving the evolution of intelligence**, and the foundation on which nature created our species—the ultimate embodiment of perceiving, learning, thinking, and doing.

Spatial intelligence plays a fundamental role in defining how we interact with the physical world. Every day, we rely on it for the most ordinary acts: parking a car by imagining the narrowing gap between bumper and curb, catching a set of keys tossed across the room, navigating a crowded sidewalk without collision, or sleepily pouring coffee into a mug without looking. In more extreme circumstances, firefighters navigate collapsing buildings through shifting smoke, making split-second judgements about stability and survival, communicating through gestures, body language and a shared professional instinct for which there's no linguistic substitute. And children spend the entirety of their pre-verbal months or years learning the world through playful interactions with their environments. All of this happens intuitively, automatically—a fluency machines have yet to achieve.

Spatial Intelligence is also foundational to our imagination and creativity. Storytellers create uniquely rich worlds in their minds and leverage many forms of visual media to bring them to others, from ancient cave painting to modern cinema to immersive video games.

Whether it's children building sandcastles on the beach or playing Minecraft on the computer, spatially-grounded imagination forms the basis for interactive experiences in real or virtual worlds. And in many industry applications, simulations of objects, scenes and dynamic interactive environments power countless numbers of critical business use cases from industrial design to digital twins to robotic training.

History is full of civilization-defining moments where spatial intelligence played central roles. In ancient Greece, Eratosthenes transformed shadows into geometry—measuring a 7-degree angle in Alexandria at the exact moment the sun cast no shadow in Syene—to calculate the Earth's circumference. Hargreave's "Spinning Jenny" revolutionized textile manufacturing through a spatial insight: arranging multiple spindles side-by-side in a single frame allowed one worker to spin multiple threads simultaneously, increasing productivity eightfold. Watson and Crick discovered DNA's structure by physically building 3D molecular models, manipulating metal plates and wire until the spatial arrangement of base pairs clicked into place. In each case, spatial intelligence drove civilization forward when scientists and inventors had to manipulate objects, visualize structures, and reason about physical spaces - none of which can be captured in text alone.

Spatial Intelligence is the scaffolding upon which our cognition is built.

It's at work when we passively observe or actively seek to create. It drives our reasoning and planning, even on the most abstract topics. And it's essential to the way we interact—verbally or physically, with our peers or with the environment itself. While most of us aren't revealing new truths on the level of Eratosthenes most days, we *routinely* think in the same way—making sense of a complex world by perceiving it through our senses, then leveraging an intuitive understanding of how it works in physical, spatial terms.

Unfortunately, today's AI doesn't think like this yet.

Tremendous progress has indeed been made in the past few years. Multimodal LLMs (MLLMs), trained with voluminous multimedia data in addition to textual data, have introduced some basics of spatial awareness, and today's AI can analyze pictures, answer questions about them, and generate hyperrealistic images and short videos. And through breakthroughs in sensors and haptics, our most advanced robots can begin to manipulate objects and tools in highly constrained environments.

Yet the candid truth is that AI's spatial capabilities remain far from human level. And the limits reveal themselves quickly. State-of-the-art MLLM models rarely perform better than chance on estimating distance, orientation, and size—or “mentally” rotating objects by regenerating them from new angles. They can't navigate mazes, recognize shortcuts, or predict basic physics. AI-generated videos—nascent and yes, very cool—often lose coherence after a few seconds.

While current state-of-the-art AI can excel at reading, writing, research, and pattern recognition in data, these same models bear fundamental limitations when representing or interacting with the physical world. Our view of the world is holistic—not just what we're looking at, but how everything relates spatially, what it means, and why it matters. Understanding this through imagination, reasoning, creation, and interaction—not just descriptions—is the power of spatial intelligence. Without it, AI is disconnected from the physical reality it seeks to understand. It cannot effectively drive our cars, guide robots in our homes and hospitals, enable entirely new ways of immersive and interactive experiences for learning and recreation, or accelerate discovery in materials science and medicine.

The philosopher Wittgenstein once wrote that “the limits of my language mean the limits of my world.” I'm not a philosopher. But I know at least for AI, there is more than just words. Spatial intelligence represents the frontier beyond language—the capability that links imagination, perception and action, and opens possibilities for machines

to truly enhance human life, from healthcare to creativity, from scientific discovery to everyday assistance.

The next decade of AI: Building truly spatially intelligent machines

So how do we build spatially-intelligent AI? What's the path to models capable of reasoning with the vision of Eratosthenes, engineering with the precision of an industrial designer, creating with the imagination of a storyteller, and interacting with their environment with the fluency of a first responder?

Building spatially intelligent AI requires something even more ambitious than LLMs: world models, a new type of generative models whose capabilities of understanding, reasoning, generation and interaction with the semantically, physically, geometrically and dynamically complex worlds - virtual or real - are far beyond the reach of today's LLMs. The field is nascent, with current methods ranging from abstract reasoning models to video generation systems. World Labs was founded in early 2024 on this conviction: that foundational approaches are still being established, making this the defining challenge of the next decade.

In this emerging field, what matters most is establishing the principles that guide development. For spatial intelligence, I define world models through **three essential capabilities**:

1. Generative: World models can generate worlds with perceptual, geometrical, and physical consistency

World models that unlock spatial understanding and reasoning must also generate simulated worlds of their own. They must be capable of spawning endlessly varied and diverse simulated worlds that follow semantic or perceptual instructions—*while* remaining geometrically, physically, and dynamically consistent—whether representing real or virtual spaces. The research community is actively exploring whether these worlds should be represented implicitly or explicitly in terms of

the innate geometric structures. Furthermore, in addition to powerful latent representations, I believe the outputs of a universal world model must also allow the generation of an explicit, observable state of the worlds for many different use cases. In particular, its understanding of the present must be tied coherently to its past; to the previous states of the world that led to the current one.

2. Multimodal: World models are multimodal by design

Just as animals and humans do, a world model should be able to process inputs—known as “prompts” in the generative AI realm—in a wide range of forms. Given partial information—whether images, videos, depth maps, text instructions, gestures, or actions—world models should predict or generate world states as *complete* as possible. This requires processing visual inputs with the fidelity of real vision while interpreting semantic instructions with equal facility. This enables both agents and humans to communicate with the model about the world through diverse inputs and receive diverse outputs in return.

3. Interactive: World models can output the next states based on input actions

Finally, if actions and/or goals are part of the prompt to a world model, its outputs must include the *next* state of the world, represented either implicitly or explicitly. When given only an action with or without a goal state as the input, the world model should produce an output consistent with the world’s previous state, the intended goal state if any, and its semantic meanings, physical laws, and dynamical behaviors. As spatially intelligent world models become more powerful and robust in their reasoning and generation capabilities, it is conceivable that in the case of a given goal, the world models themselves would be able to predict not only the next state of the world, but also the next actions based on the new state.

The scope of this challenge exceeds anything AI has faced before.

While language is a purely generative phenomenon of human cognition, worlds play by much more complex rules. Here on Earth, for instance, gravity governs motion, atomic structures determine how light produces colors and brightness, and countless physical laws constrain every interaction. Even the most fanciful, creative worlds are composed of spatial objects and agents that obey the physical laws and dynamical behaviors that define them. Reconciling all of this consistently—the semantic, the geometric, the dynamic, and physical—demands entirely new approaches. The dimensionality of representing a world is vastly more complex than that of a one-dimensional, sequential signal like language. Achieving world models that deliver the kind of universal capabilities we enjoy as humans will require overcoming several formidable technical barriers. At World Labs, our research teams are devoted to making fundamental progress toward that goal.

Here are some examples of our current research topics:

- **A new, universal task function for training:** Defining a universal task function as simple and elegant as next-token prediction in LLMs has long been a central goal of world model research. The complexities of both their input and output spaces make such a function inherently more difficult to formulate. But while much remains to be explored, this objective function and corresponding representations must reflect the laws of geometry and physics, honoring the fundamental nature of world models as grounded representations of both imagination and reality.

- **Large-scale training data:** Training world models requires far more complex data than text curation. The promising news: massive data sources already exist. Internet-scale collections of images and videos represent abundant, accessible training material—the challenge lies in developing algorithms that can extract deeper spatial information from these two-dimensional image or video frame-based signals (i.e. RGB). Research over the past decade has shown the power of scaling laws linking data volume and model size in language models; the key unlock for world models is building architectures that can leverage existing visual data at comparable scale. In addition, I would not underestimate the power of high-quality synthetic data and additional modalities like depth and tactile information. They supplement the internet scale data in critical steps of the training process. But the path forward depends on better sensor systems, more robust signal extraction algorithms, and far more powerful neural simulation methods.
- **New model architecture and representational learning:** World model research will inevitably drive advances in model architecture and learning algorithms, particularly beyond the current MLLM and video diffusion paradigms. Both of these typically tokenize data into 1D or 2D sequences, which makes simple spatial tasks unnecessarily difficult - like counting unique chairs in a short video, or remembering what a room looked like an hour ago. Alternative architectures may help, such as 3D or 4D-aware methods for tokenization, context, and memory. For example, at World Labs, our recent work on a real-time generative frame-based model called RTFM has demonstrated this shift, which uses spatially-grounded frames as a form of spatial memory to achieve efficient real-time generation while maintaining persistence in the generated world.

Clearly, we are still facing daunting challenges before we can fully unlock spatial intelligence through world modeling. This research isn't just a theoretical exercise. It is the core engine for a new class of creative and productivity tools. And the progress within World Labs has

been encouraging. We recently shared with a limited number of users a glimpse of Marble, the first ever world model that can be prompted by multimodal inputs to generate and maintain consistent 3D environments for users and storytellers to explore, interact with, and build further in their creative workflow. And we are working hard to make it available to the public soon!

Marble is only our first step in creating a truly spatially intelligent world model. As the progress accelerates, researchers, engineers, users, and business leaders alike are beginning to recognize its extraordinary potential. The next generation of world models will enable machines to achieve spatial intelligence on an entirely new level—an achievement that will unlock essential capabilities still largely absent from today's AI systems.

Using world models to build a better world for people

It matters what motivates the development of AI. As one of the scientists who helped usher in the era of modern AI, my motivation has always been clear: AI must augment human capability, not replace it. For years, I've worked to align AI development, deployment, and governance with human needs. Extreme narratives of techno-utopia and apocalypse are abundant these days, but I continue to hold a more pragmatic view: AI is developed by people, used by people, and governed by people. It must always respect the agency and dignity of people. Its magic lies in extending our capabilities; making us more creative, connected, productive, and fulfilled. Spatial intelligence represents this vision—AI that empowers human creators, caregivers, scientists, and dreamers to achieve what was once impossible. This belief is what drives my commitment to spatial intelligence as AI's next great frontier.

The applications of spatial intelligence span varying timelines. Creative tools are emerging now—World Labs' Marble already puts these

capabilities in creators' and storytellers' hands. Robotics represents an ambitious mid-term horizon as we refine the loop between perception and action. The most transformative scientific applications will take longer but promise a profound impact on human flourishing.

Across all these timelines, several domains stand out for their potential to reshape human capability. It will take significant collective effort, more than a single team or a company can possibly achieve. It will require participation across the entire AI ecosystem—researchers, innovators, entrepreneurs, companies, and even policymakers—working toward a shared vision. But this vision is worth pursuing. Here's what that future holds:

Creativity: Superpowering storytelling and immersive experiences

"Creativity is intelligence having fun." This is one of my favorite quotes by my personal hero Albert Einstein. Long before written language, humans told stories—painted them on cave walls, passed them through generations, built entire cultures on shared narratives. Stories are how we make sense of the world, connect across distance and time, explore what it means to be human, and most importantly, find meaning in life and love within ourselves. Today, spatial intelligence has the potential to transform how we create and experience narratives in ways that honor their fundamental importance, and extend their impacts from entertainment to education, from design to construction.

World Labs' Marble platform will be putting unprecedented spatial capabilities and editorial controllability in the hands of filmmakers, game designers, architects, and storytellers of all kinds, allowing them to rapidly create and iterate on fully explorable 3D worlds without the overhead of conventional 3D design software. The creative act remains as vital and human as ever; the AI tools simply amplify and accelerate what creators can achieve. This includes:

- Narrative experiences in new dimensions: Filmmakers and game designers are using Marble to conjure entire worlds without the constraints of budget or geography, exploring varieties of scenes and perspectives that would have been intractable to explore within a traditional production pipeline. As the lines between different forms of media and entertainment blur, we're approaching fundamentally new kinds of interactive experiences that blend art, simulation, and play—personalized worlds where anyone, not just studios, can create and inhabit their own stories. With the rise of newer, more rapid ways to lift concepts and storyboards into full experiences, narratives will no longer be bound to a single medium, with creators free to build worlds with shared throughlines across myriad surfaces and platforms.
- Spatial narratives through design: Essentially every manufactured object or constructed space must be designed in virtual 3D before its physical creation. This process is highly iterative and costly in terms of both time and money. With spatially intelligent models at their disposal, architects can quickly visualize structures before investing months into designs, walking through spaces that don't yet exist—essentially telling stories about how we might live, work, and gather. Industrial and fashion designers can translate imagination into form instantly, exploring how objects interact with human bodies and spaces.

- New immersive and interactive experiences: Experience itself is one of the deepest ways that we, as a species, create meaning. For the entirety of human history, there has been one singular 3D world: the physical one we all share. Only in recent decades, through gaming and early virtual reality (VR), have we begun to glimpse what it means to share alternate worlds of our own creation. Now, spatial intelligence combined with new form factors, like VR and extended reality (XR) headsets and immersive displays, elevates these experiences in unprecedented ways. We're approaching a future where stepping into fully realized multi-dimensional worlds becomes as natural as opening a book. Spatial intelligence makes world-building accessible not just to studios with professional production teams but to individual creators, educators, and anyone with a vision to share.

Robotics: Embodied intelligence in action

Animals from insects to humans depend on spatial intelligence to understand, navigate and interact with their worlds. Robots will be no different. Spatially-aware machines have been the dream of the field since its inception, including my own work with my students and collaborators at my Stanford research lab. This is also why I'm so excited by the possibility of bringing them about using the kinds of models World Labs is building.

- **Scaling robotic learning via world models:** The progress of robotic learning hinges on a scalable solution of viable training data. Given the enormous state spaces of possibilities that robots have to learn to understand, reason, plan, and interact with, many have conjectured that a combination of internet data, synthetic simulation, and real-world capture of human demonstration are required to truly create generalizable robots. But unlike language models, training data is scarce for today's robotic research. World models will play a defining role in this. As they increase their perceptual fidelity and computational efficiency, outputs of world models can rapidly close the gap between simulation and reality. This will in turn help train robots across simulations of countless states, interactions and environments.
- **Companions and collaborators:** Robots as human collaborators, whether aiding scientists at the lab bench or assisting seniors living alone, can expand part of the workforce in dire need of more labour and productivity. But doing so demands spatial intelligence that perceives, reasons, plans, and acts while—and this is most important—staying empathetically aligned with human goals and behaviors. For instance, a lab robot might handle instruments so the scientist can focus on tasks needing dexterity or reasoning, while a home assistant might help an elderly person cook without diminishing their joy or autonomy. Truly spatially intelligent world models that can predict the next state or possibly even actions consistent with this expectation are critical for achieving this goal.

- **Expanding forms of embodiment:** Humanoid robots play a role in the world we've built for ourselves. But the full benefit of innovation will come from a far more diverse range of designs: nanobots that deliver medicine, soft robots that navigate tight spaces, and machines built for the deep sea or outer space. Whatever their form, future spatial intelligence models must integrate both the environments these robots inhabit and their own embodied perception and movement. But a key challenge in developing these robots is the lack of training data in these wide varieties of embodied form factors. World models will play a critical role in simulation data, training environments, and benchmarking tasks for these efforts.

The Longer Horizon: Science, Healthcare, and Education

In addition to creative and robotics applications, spatial intelligence' profound impact will also extend to fields where AI can enhance human capability in ways that save lives and accelerate discovery. I highlight below three areas of applications that can be deeply transformative, though it goes without saying the use cases of spatial intelligence are truly expansive across many more industries.

In **scientific research**, spatially intelligent systems can simulate experiments, test hypotheses in parallel, and explore environments inaccessible to humans—from deep oceans to distant planets. This technology can transform computational modeling in fields like climate science and materials research. By integrating multi-dimensional simulation with real-world data collection, these tools can lower compute barriers and extend what every laboratory can observe and understand.

In **healthcare**, spatial intelligence will reshape everything from laboratory to bedside. At Stanford, my students and collaborators have spent many years working with hospitals, elder care facilities, and patients at home. This experience has convinced me of spatial

intelligence's transformative potential here. AI can accelerate drug discovery by modeling molecular interactions in multi-dimensions, enhance diagnostics by helping radiologists spot patterns in medical imaging, and enable ambient monitoring systems that support patients and caregivers without replacing the human connection that healing requires, not to mention the potential of robots in helping our healthcare workers and patients in many different settings.

In **education**, spatial intelligence can enable immersive learning that makes abstract or complex concepts tangible, and create iterative experiences so essential to how our brains and bodies are wired in learning. In the age of AI, the need for faster and more effective learning and reskilling is particularly important for both school-aged children and adults. Students can explore cellular machinery or walk through historical events in multi-dimentionality. Teachers gain tools to personalize instruction through interactive environments. Professionals—from surgeons to engineers—can safely practice complex skills in realistic simulations.

Across all these domains, the possibilities are boundless, but the goal remains constant: AI that augments human expertise, accelerates human discovery, and amplifies human care—not replacing the judgment, creativity, and empathy that are central for being humans.

Conclusion

The last decade has seen AI become a global phenomenon and an inflection point in technology, the economy, and even geopolitics. But as a researcher, educator, and now, entrepreneur, it's still the spirit behind Turing's 75-year-old question that inspires me most. I still share his sense of wonder. It's what energizes me every day by the challenge of spatial intelligence.

For the first time in history, we're poised to build machines so in tune with the physical world that we can rely on them as true partners in the

greatest challenges we face. Whether accelerating how we understand diseases in the lab, revolutionizing how we tell stories, or supporting us in our most vulnerable moments due to sickness, injury, or age, we're on the cusp of technology that elevates the aspects of life we care about most. This is a vision of deeper, richer, more empowered lives.

Almost a half billion years after nature unleashed the first glimmers of spatial intelligence in the ancestral animals, we're lucky enough to find ourselves among the generation of technologists who may soon endow machines with the same capability—and privileged enough to harness those capabilities for the benefits of people everywhere. Our dreams of truly intelligent machines will not be complete without spatial intelligence.

This quest is my North Star. [Join me](#) in pursuing it.

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